PYTHON PROGRAMMING



21 Days Challenge

Day-13

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PYTHON OOP FUNDAMENTALS: CLASSES, OBJECTS, ATTRIBUTES, AND METHODS



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Class

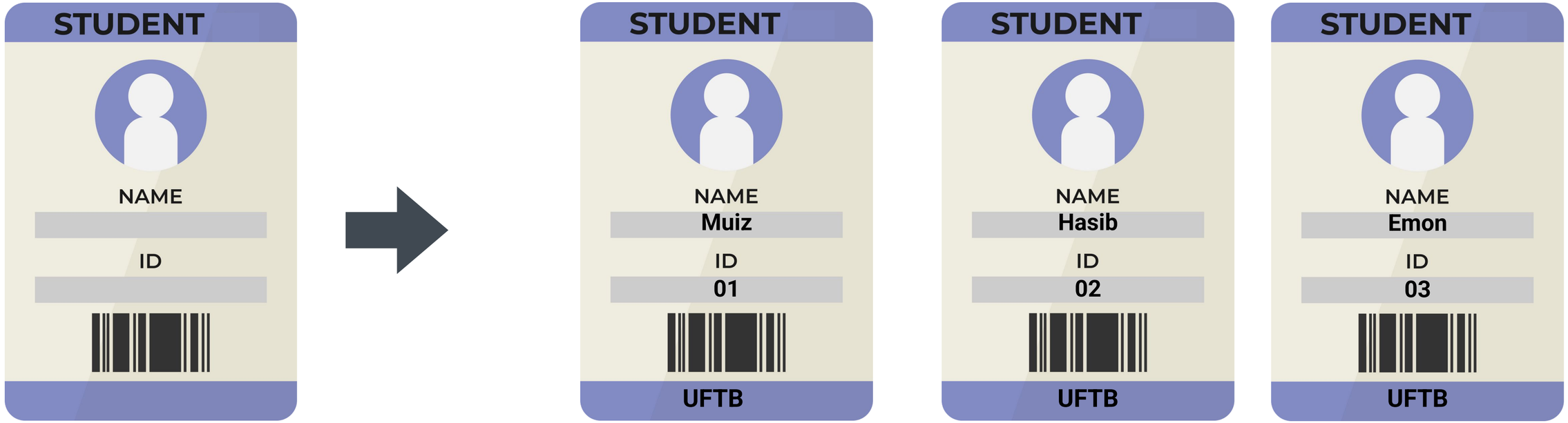
A class in Python is a user-defined template for creating objects.

```
class Student:
    pass
```



Object

An object is a specific instance of a class.

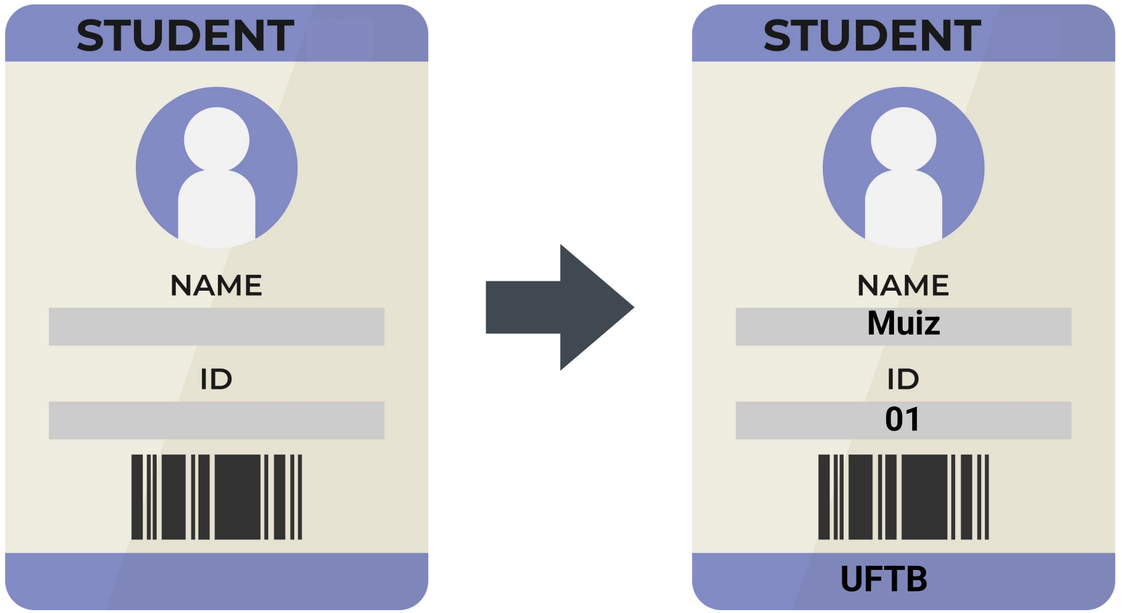


Object (Contd...)

An object is a specific instance of a class. It holds its own set of data.

```
class Student:
    pass

Student1 = Student()
Student2 = Student()
```

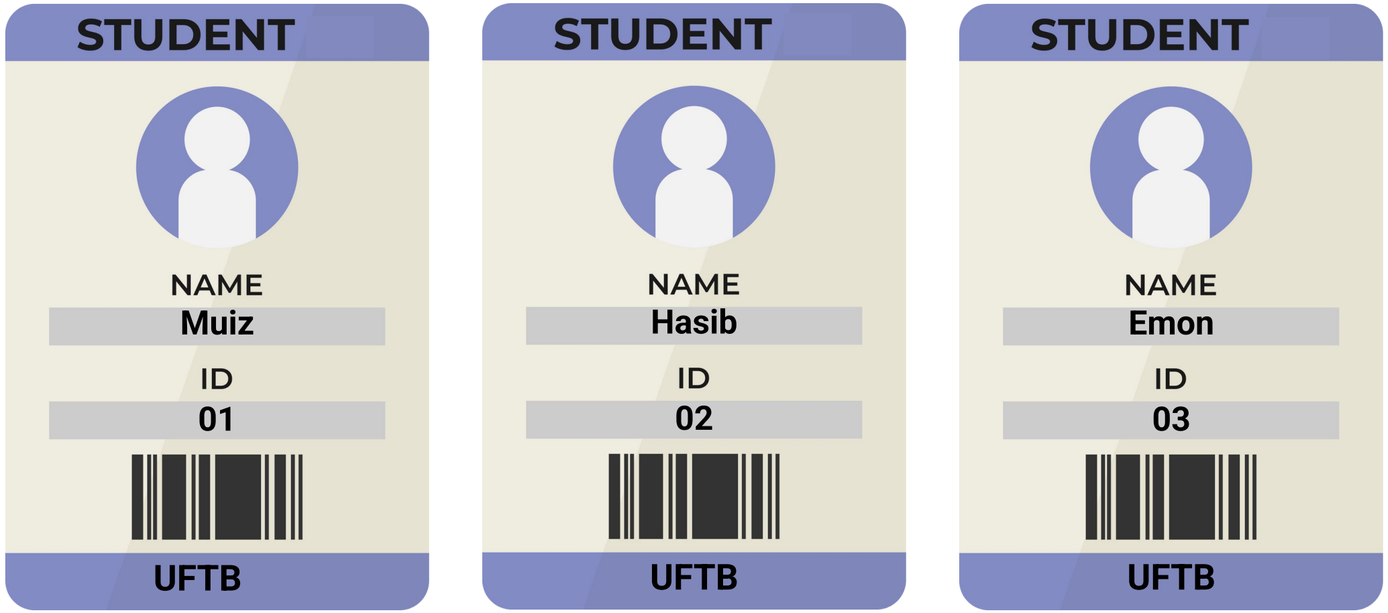


Class Attributes

Class attributes are shared by all students.

```
class Student:
    Institution = "UFTB"

Student1 = Student()
Student2 = Student()
```



Constructor

The constructor runs automatically when we create a student.

```
class Student:
    def __init__(self):
        pass

Student1 = Student()
Student2 = Student()
```

Instance Attributes

Instance attributes belong to each object separately.

```
class Student:
    def __init__(self, name, student_id):
        self.name = name
        self.student_id = student_id

    Student2 = Student("Emon",03)


    Student2.name
    Student2.student_id
```



Instance Methods

```
class Student:
    def __init__(self, name, student_id):
        self.name = name
        self.student_id = student_id

    def display_info(self):
        print(f"Name: {self.name}, ID: {self.student_id}")
Student1 = Student()
Student1.display_info()
```

Class Methods

```
class Student:
    total_students = 0
    def __init__(self, name, student_id):
        self.name = name
        self.student_id = student_id
        Student.total_students += 1

    @classmethod
    def get_total(cls):
        return cls.total_students
Student.get_total_students()
```

Static Method

```
class Student:
    @staticmethod
    def school_motto():# no use self,cls
        return "Knowledge is Power!"


    Student.school_motto()) # use in class


    s1=Student()
    s1.school_motto()   # use in instance
```

Summery

Instance attributes → name, ID

Instance method → display_info() shows details

Class attribute → all students belong to "UFTB"

Class method → total_students() counts enrolled students

Static method → school_motto() same for everyone

QUESTION SESSION

THANK YOU

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